(ii) In the right angled ΔGOH,

$$\sin 24^\circ = \frac{OH}{GH}$$

$$0.4067 = \frac{OH}{250} \; ; \; OH = 101.68$$

Distance of H to the East of $G = 101.68$ km

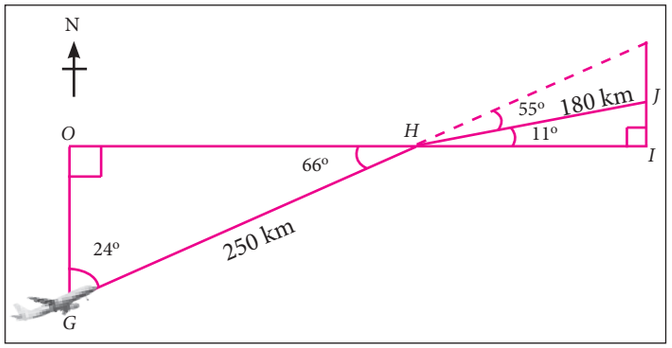


Fig. 6.18 (b)

(iii) In the right angled ΔHIJ,

$$\sin 11^\circ = \frac{IJ}{HJ}$$

$$0.1908 = \frac{IJ}{180} \; ; \; IJ = 34.34 \text{ km}$$

Distance of J to the North of $H = 34.34$ km

(iv) In the right angled ΔHIJ,

$$\cos 11^\circ = \frac{HI}{HJ}$$

$$0.9816 = \frac{HI}{180} \; ; \; HI = 176.69 \text{ km}$$

Distance of J to the East of $H = 176.69$ km

Example 6.25 As shown in the figure, two trees are standing on flat ground. The angle of elevation of the top of both the trees from a point X on the ground is 40°. If the horizontal distance between X and the smaller tree is 8 m and the distance of the top of the two trees is 20 m, calculate

(i) the distance between the point X and the top of the smaller tree.

(ii) the horizontal distance between the two trees.
$(\cos 40^\circ = 0.7660)$

Solution Let AB be the height of the bigger tree and CD be the height of the smaller tree and X is the point on the ground.

(i) In the right angled ΔXCD, $\cos 40^\circ = \frac{CX}{XD}$

$$XD = \frac{8}{0.7660} = 10.44 \text{ m}$$

Therefore, the distance between X and top of the smaller tree = XD = 10.44 m

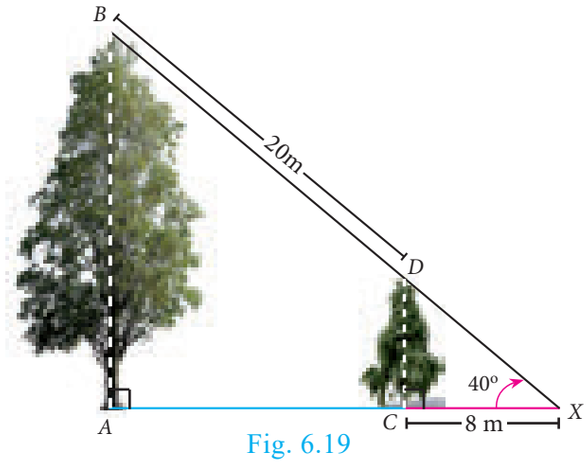


Fig. 6.19

(ii) In the right angled ΔXAB,

$$\cos 40^\circ = \frac{AX}{BX} = \frac{AC + CX}{BD + DX}$$

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